

NATIONAL UNIVERSITY OF SINGAPORE
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

ACADEMIC YEAR 2025-2026
SEMESTER 1

EE2213: Introduction to AI

TUTORIAL 2: L2-L3

Question 1 (Lecture 2)

Which of the following statements about tree search and graph search are correct? Select all that apply.

- (a) If a state is reachable via multiple paths, tree search will create two separate nodes corresponding to each path
- (b) Tree search checks if a node has been visited before adding it to the search queue
- (c) Tree search is more efficient than graph search
- (d) Graph search checks if a node has been visited before adding it to the search queue

Question 2 (Lecture 3)

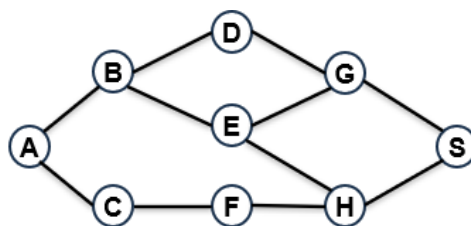
Which of the following is true about greedy best-first search? Select all that apply.

- (a) It guarantees both completeness and optimality when using a well-designed heuristic
- (b) It expands nodes based solely on the lowest cumulative path cost from the start state
- (c) It expands the node that appears closest to the goal, based only on the heuristic function
- (d) It behaves identically to uniform-cost search when edge costs are equal

Question 3 (Lecture 2)

You are given the following undirected graph.

- (i) Write a Python program to perform a Breadth-First Search (BFS) traversal starting from node 'A'. Print the order in which nodes are visited.
- (ii) Write a Python program to perform a Depth-First Search (DFS) traversal starting from node 'A'. Print the order in which nodes are visited.

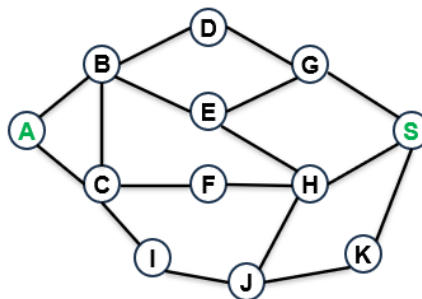


Question 4 (Lecture 2)

Your friend is flying from Arequipa (A), a historic city and the second-largest in Peru, to Singapore (S). However, there is no direct flight between the two cities.

You decide to plan a route with the minimum number of flight segments needed for your friend. To do this, you model the available flight connections as an undirected graph as shown below, where nodes represent cities, and edges represent direct flights between cities. The modeled graph is shown below.

- (i) Which strategy is better for finding the fewest number of flight segments: Breadth-First Search (BFS) or Depth-First Search (DFS)?
- (ii) Write a Python programme to compute the minimum number of flight segments from Arequipa to Singapore.



Question 5 (Lecture 3)

Signal Propagation in a Network (Adapted from LeetCode #743)

You are given a communication network consisting of several nodes connected by directed links, each with an associated transmission delay (indicated by the numbers beside each edge), as illustrated in the graph below. Given a starting node *a* (the leftmost node), determine the minimum time required for a signal to reach all nodes in the network.

- (i) Which search algorithm that you have learned so far would be most suitable for this problem?
- (ii) Implement your solution in Python using the selected search algorithm. Your program should:
 - Model the network as a graph.
 - Compute the minimum time it takes for the signal to reach every node.

